Lab. 2

Linear Op-Amp Circuits

Team No. 2

Names:

Jefferson Chinchilla Quesada	2023152266
Mattio Coghi Quirós	2023
Nicolás Mena Valerio	2022

Reference:

Floyd Thomas L., Buchla David M.: *Experiments in Analog Fundamentals, A System Approach*. Pearson Education, 2013.

Reading:

Floyd and Buchla, *Fundamentals: A Systems Approach*, Sections 6-4 through 6-6. Pearson Education, 2013.

Objectives:

After performing this experiment, you will be able to:

1. Construct and test inverting and noninverting amplifiers using op-amps.
2. Specify components for inverting and noninverting amplifiers using op-amps.

Materials Needed:

- Resistors: two 1 kΩ, one 10 kΩ, one 470 kΩ, one 1.0 MΩ
- Two 1.0 μF capacitors
- One 741C op-amp
- *For Further Investigation:*
  - One 1.0 kΩ potentiometer, one 100 kΩ resistor, assorted resistors to test.

Summary of Theory

One of the most important ideas in electronics incorporates the idea of *feedback*, where a portion of the output is returned to the input. If the return signal tends to decrease the input amplitude, it is called *negative feedback*. Negative feedback produces a number of desirable qualities in an amplifier, increasing its stability and its frequency response. It also allows the gain to be controlled independently of the device parameters, temperature, or other variables.

Operational amplifiers are almost always used with external, negative feedback. The feedback circuit determines the specific characteristics of the amplifier. By itself, an op-amp has an extremely high voltage gain called the *open-loop* gain, A_{ol}. When negative feedback is added, the overall gain of the amplifier is determined by the feedback circuit. This gain, including the feedback circuit, is called the *closed-loop* gain, A_{cl}.

Figure 1 illustrates a noninverting amplifier with negative feedback. The input is applied to the noninverting terminal and a fraction (B) of the output is returned to the inverting input by the voltage divider. The very high gain of the amplifier forces the two inputs to be very nearly the same voltage; therefore, the input voltage is across R_i, and the output voltage is across $R_i + R_f$. The closed-loop gain of this noninverting amplifier is given as $A_{cl(NI)}$. The closed-loop gain is the reciprocal of the feedback fraction, as shown in Figure 1.

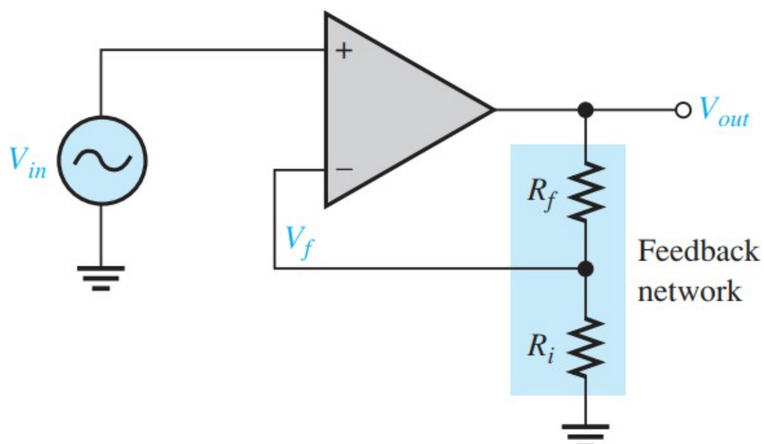

Figure 1: *Noninverting amplifier*

$$\text{feedback fraction} = B = \frac{R_i}{R_i + R_f}$$

$$A_{cl(NI)} = \frac{1}{B} = \frac{R_i + R_f}{R_i} = 1 + \frac{R_f}{R_i}$$

An inverting amplifier is shown in Figure 2. In this amplifier, the output voltage is the opposite phase to the input voltage. The noninverting input is grounded, and the input signal is applied through a resistor to the inverting terminal. A feedback resistor is connected between the output and the inverting input. This amplifier can be analyzed by assuming the input current is the same as the current in the feedback resistor, $I_{in} = I_f$, and that the open-loop gain of the amplifier is very high. From these assumptions, the closed-loop gain can be calculated quite accurately as the ratio of the feedback resistor to the input resistor. The basic equations for the inverting amplifier are shown with Figure 2. The minus sign indicates inversion between the input and output. (−) refers to the inverting input on the op-amp.

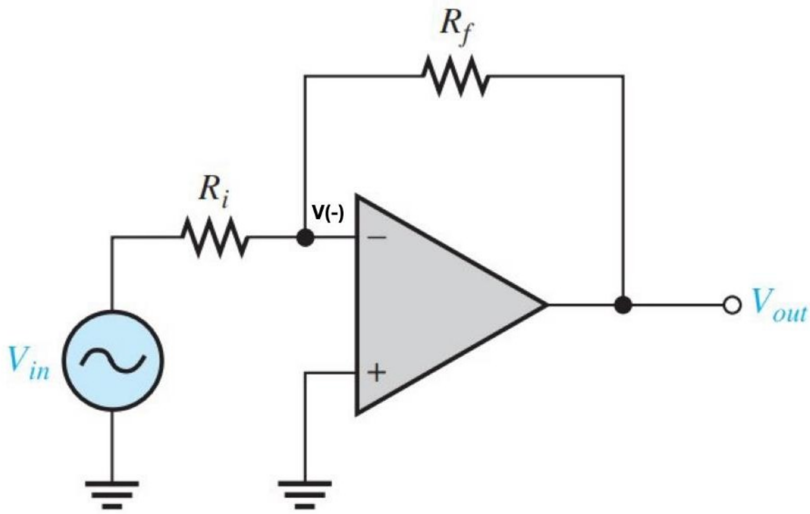

Figure 2: *Inverting amplifier*

$$I_{in} = I_f$$

A_{ol} is very large, therefore

$$V(-) = -\frac{V_{out}}{A_{ol}} \cong 0$$

$$\frac{V_{in}}{R_i} = -\frac{V_{out}}{R_f}$$

$$A_{cl(I)} = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

Procedure

1. The circuit to be tested in this step is the noninverting amplifier illustrated in Figure 3.

   a. Measure a $10\,\text{k}\Omega$ resistor for R_f and $1.0\,\text{k}\Omega$ resistor for R_i. Record the measured value of resistance in Table 1.

   b. Using the measured resistances, compute the closed-loop gain of the noninverting amplifier. The closed-loop gain equation is given next to Figure 1.

   c. Calculate V_{out} using the computed closed-loop gain.

   d. Connect the circuit shown in Figure 3. Set the signal generator for a $500\,\text{mV}_{pp}$ sinusoidal wave at 1.0 kHz. The generator should have no dc offset.

   e. Measure the output voltage, V_{out}. Record the measured value.

   f. Measure the feedback voltage at pin 2. Record the measured value.

   g. Place a $1.0\,\text{M}\Omega$ test resistor in series with the generator. Measure the input resistance of the circuit, R_{in}, based on the voltage drop across the test resistor. Use the voltage divider rule to indirectly find the input resistance.

Table 1: *Noninverting amplifier measurements*

R_f Measured	R_i Measured	V_{in} Measured	$A_{cl(NI)}$ Computed	V_{out} Computed (pin 6)	V_{out} Measured (pin 6)	$V(-)$ Measured (pin 2)	R_{in} Measured
10 kΩ	1 kΩ	$500\,\text{mV}_{pp}$	11	5.5 Vpp	5.56 Vpp	556 mV	654 kΩ

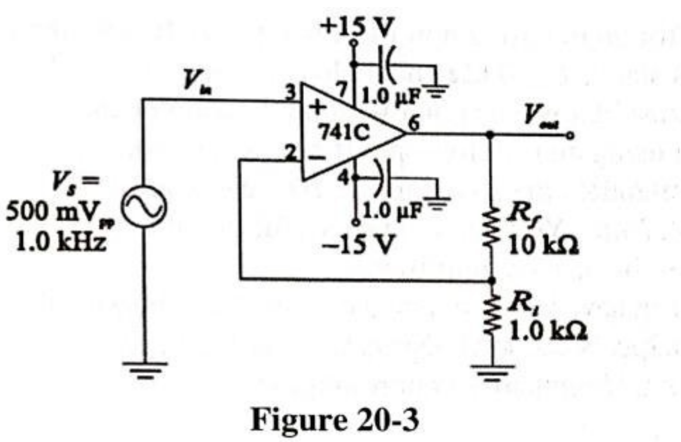

Figure 3: *Noninverting amplifier test circuit (741C, $R_f = 10\,k\Omega$, $R_i = 1.0\,k\Omega$)*

El R_{in} de la tabla se obtiene con la regla del divisor de tensión. El generador de $500\,\text{mV}_{pp}$ equivale a $176.8\,\text{mV}_{rms}$ y en la entrada del amplificador se midieron $69.91\,\text{mV}_{rms}$, así que sobre el resistor de prueba de $1.0\,\text{M}\Omega$ caen los $106.87\,\text{mV}_{rms}$ restantes.

$$R_{in} = R_{prueba}\frac{V_{R_{in}}}{V_s - V_{R_{in}}} = 1\,\text{M}\Omega \cdot \frac{69.91\,\text{mV}}{106.87\,\text{mV}} = 654\,\text{k}\Omega$$

Este valor queda muy por debajo de lo que predice la teoría, porque la realimentación negativa multiplica la resistencia de entrada del 741C por $(1 + A_{ol}B)$ y la lleva al orden de los GΩ. Lo que limita la medición es la carga del instrumento, la punta del osciloscopio y el camino de corriente de polarización quedan en paralelo con la entrada y son del orden de los MΩ, de modo que el divisor termina midiendo esa combinación.

2. In this step you will test an inverting amplifier. All data are to be recorded in Table 2. The circuit is illustrated in Figure 4. The closed-loop gain is:

$$A_{cl(I)} = -\frac{R_f}{R_i}$$

   a. Use the same resistors for R_f and R_i as in step 1. Record the measured value of resistance in Table 2.
   b. Using the measured resistance, compute and record the closed-loop gain of the inverting amplifier.
   c. Calculate the V_{out} using the computed closed-loop gain.
   d. Connect the circuit shown in Figure 4. Set the signal generator for the $500\,\mathrm{mV}_{pp}$ sine wave at 1 kHz. The generator should have no dc offset.
   e. Measure and record the output voltage, V_{out}.
   f. Measure and record the output voltage at pin 2. This point is called a *virtual ground* because of the effect of negative feedback.
   g. Place a $1.0\,\mathrm{k\Omega}$ test resistor in series with the generator and R_i. Measure the input resistance of the circuit, R_{in}, based on the voltage drop across the test resistor. In this case, you should observe that the drop across the test resistor is about the same as the drop across R_i.

Table 2: *Inverting amplifier measurements*

R_f Measured	R_i Measured	V_{in} Measured	$A_{cl(NI)}$ Computed	V_{out} Computed (pin 6)	V_{out} Measured (pin 6)	$V(-)$ Measured (pin 2)	R_{in} Measured
10 kΩ	1 kΩ	$50\,\mathrm{mV}_{pp}$	−10	5 V	4.48 V	4.39 mV	976 Ω

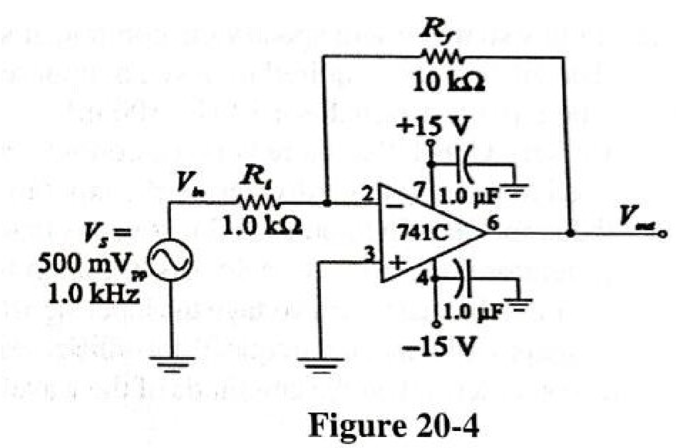

Figure 4: *Inverting amplifier test circuit (741C, $R_f = 10\,\mathrm{k\Omega}$, $R_i = 1.0\,\mathrm{k\Omega}$)*

Acá el resistor de prueba es de $1.0\,\mathrm{k\Omega}$ y el nodo de entrada quedó en $87.29\,\mathrm{mV}_{rms}$, prácticamente la mitad de los $176.8\,\mathrm{mV}_{rms}$ del generador, tal como anticipa el enunciado.

$$R_{in} = 1\,\mathrm{k\Omega} \cdot \frac{87.29\,\mathrm{mV}}{176.78\,\mathrm{mV} - 87.29\,\mathrm{mV}} = 976\,\Omega$$

El resultado cae a un 1.4 % del R_i medido de 0.9897 kΩ. Tiene sentido, en el amplificador inversor la señal entra por R_i contra la tierra virtual del pin 2, así que la resistencia de entrada del circuito queda fijada por ese resistor.

3. In this step you will specify the components for an inverting amplifier using a 741C Op-Amp. The amplifier is required to have an input resistance of $10\,\mathrm{k}\Omega$ and a closed-loop gain of -47. The input test signal is a $1\,\mathrm{kHz}$, $100\,\mathrm{mV}_{pp}$ sinusoidal wave signal with no dc component (offset). Check that there is no dc component using an oscilloscope. If necessary, you may need to put a voltage divider on the input to attenuate the signal to the $100\,\mathrm{mV}$ level.

   Draw the amplifier. Then build and test your circuit.

   Note: You need to be careful that the generator does not have a dc offset: remember this is a dc amplifier!

   Find the maximum voltage the input signal can have before clipping occurs. Try increasing the frequency and note the frequency at which the output is distorted. Does the upper frequency response depend on the amplitude of the waveform? Summarize your results in the space provided.

A continuación se detalla el proceso de simulación de esta sección:

Primero se calcula el valor de R_f mediante una simple ecuación:

$$A_{cl} = -\frac{R_f}{R_i}$$

$$47 = \frac{R_f}{10\ \mathrm{k}\Omega}$$

$$R_f = 470\ \mathrm{k}\Omega$$

Ahora para calcular en que punto el amplificador se satura basta con hacer la siguiente ecuación:

$$15\ \mathrm{V} = -47 \cdot V_{in_{max}}$$

$$V_{in_{max}} = 320\ \mathrm{mV}$$

Este comportamiento se evidencia en las siguientes capturas:

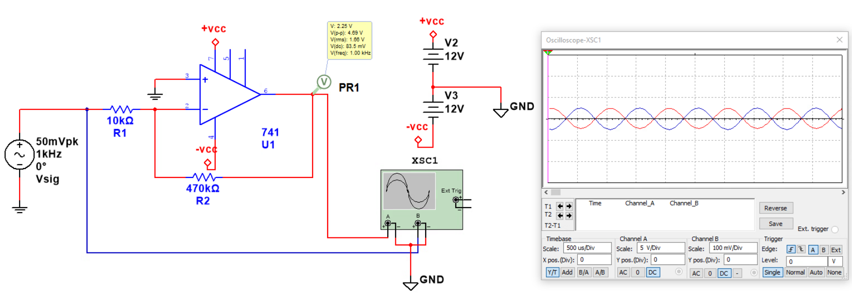

Figure 5: *Vin de* 100 *mVpp*

En la figura 7 se observa como la señal de salida se empieza a recortar, ya que supera los 15 V.

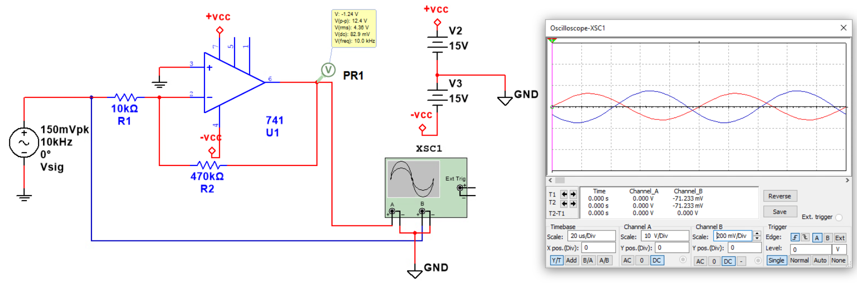

Figure 6: *Vin de* 300 *mVpp*

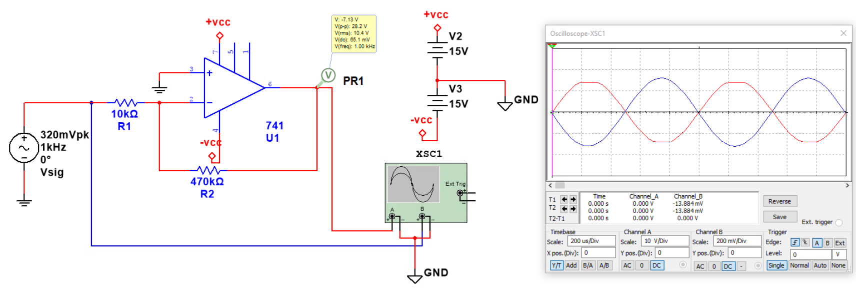

Figure 7: *Vin de* 640 *mVpp*

En cuanto a las simulaciones de respuesta en frecuencia se utilizó un barrido AC para comprobar el comportamiento del circuito a medida que se aumentaba la frecuencia de la señal de entrega. El resultado de esta simulación se muestra en la figura 8.

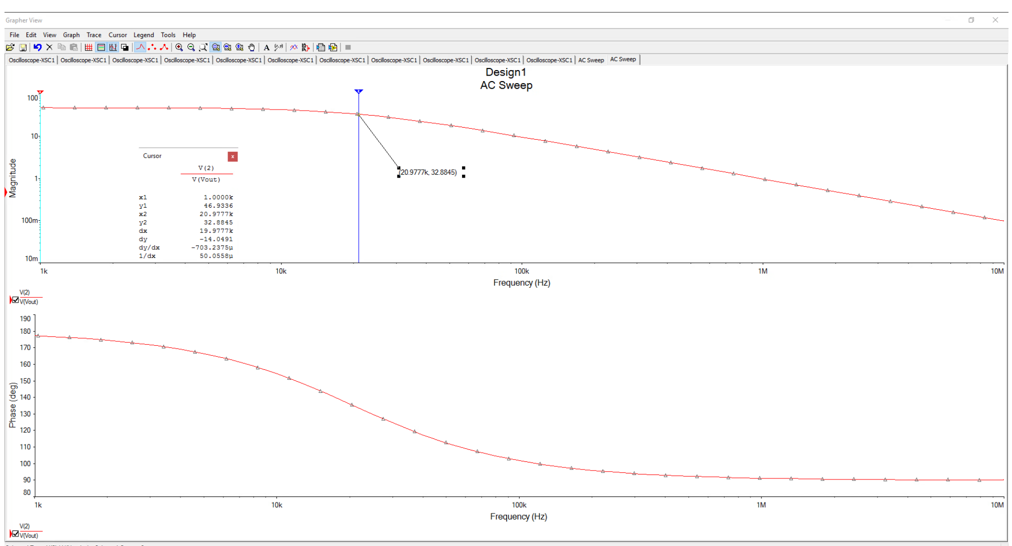

Figure 8: *Respuesta en frecuencia del amplificador Inversor*

En dicha imagen se observa el punto exacto en donde el circuito deja de realizar su función de amplificación, mostrando claramente la frecuencia de corte ($f_c = 21$ kHz). Sin embargo, la señal no se distorsiona con la frecuencia, sino que se atenuá. Para determinar la dependencia de la frecuencia y la magnitud, se probaron valores de V_{in} y f hasta alcanzar una distorsión. En este caso se logro al aumentar la tensión a 600 mVp y con una frecuencia de 40 kHz, este resultado se observa en la figura 9.

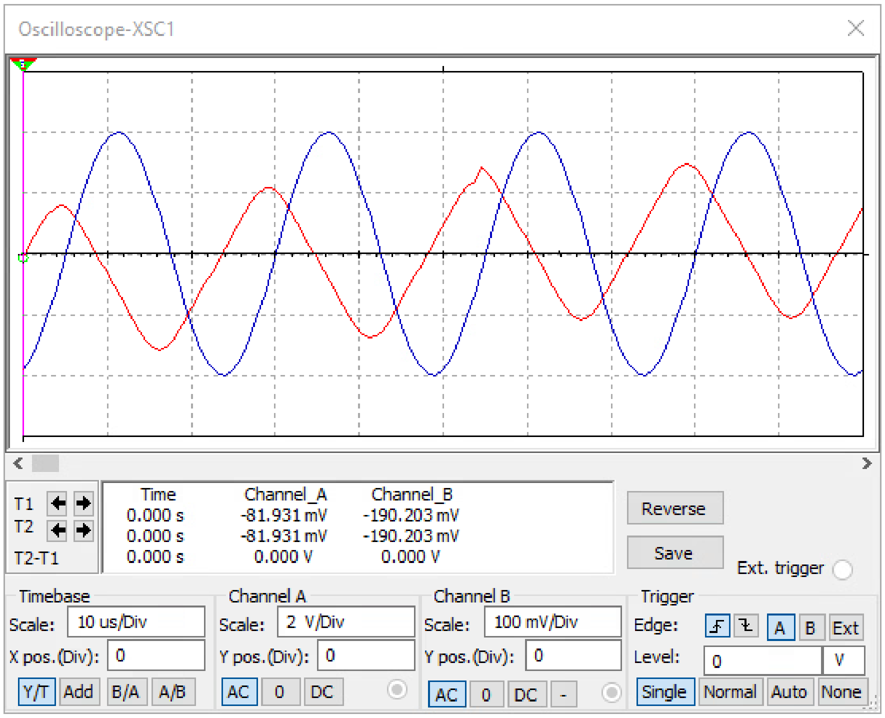

Figure 9: *Distorsión de la señal*

Conclusion

Evaluation and Review Questions

1. Express the gain of the amplifiers tested in steps 1 and 2 in dB.

$$A_{cl(NI)} = 20log(11) = 20.8 \text{ dB}$$
$$A_{cl(I)} = 20log(10) = 20 \text{ dB}$$

2. It was correct to talk about a virtual ground for the inverting amplifier. Why isn't it correct to refer to a virtual ground for the noninverting amplifier?

   Debido a la configuración de No inversor la terminal negativa no está conectada a tierra.

3. If $R_f = R_i = 10\,\text{k}\Omega$, what gain would you expect for:

   a. A noninverting amplifier?

   $$A_{cl(NI)} = 1 + \frac{10 \text{ k}\Omega}{10 \text{ k}\Omega} = 2$$

   b. An inverting amplifier?

   $$A_{cl(I)} = \frac{-R_f}{R_i} = -1$$

4. For the noninverting amplifier in Figure 20-3:

   a. If $R_f = 0$ and R_i is infinite, what is the gain?
   Si se realizan dichos cambios, la ganancia sería 1.

   $$A_{cl(NI)} = 1 + \frac{0}{\infty} = 1$$

   b. What is this type of amplifier called?
   Corresponde a un seguidor de voltaje.

5. What output would you expect in the inverting amplifier of Figure 20-4 if R_f were open?
   La salida se recortaría, ya que la señal se amplificaría en un factor de A_{ol} el cual suele ser muy alto, provocando que el amplificador se sature. La señal de salida se vería con varios recorte en los picos y valles, pareciendo una onda cuadrada.

For Further Investigation

An interesting application of an inverting amplifier is to use it as the basis of an ohmmeter for high value resistors. The circuit is shown in Figure 10. The unknown resistor, labeled R_x, is placed between the terminals. The output voltage is proportional to the unknown resistance.

Calibrate the meter by placing a known 10 kΩ resistor in place of R_x and adjusting the potentiometer for exactly 100 mV output. The output then represents 10 mV/1000 Ω. By reading the output voltage, and moving the decimal point, you can directly read resistors from several thousand ohms to over 1 MΩ.

Construct the circuit and test it using different resistors. Calibrate output voltage against resistance and compare with theory. Find the percent error for a 1 MΩ resistor using a lab meter as a standard.

Summarize your results in a short report.

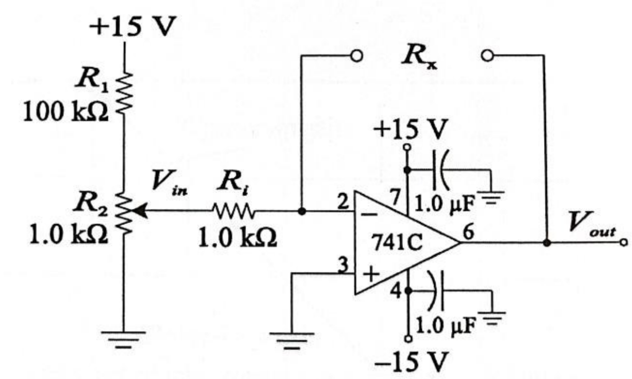

Figure 10: Ohmmeter circuit based on the inverting amplifier

A continuación se resumen los resultados de esta sección.

Con la calibración que pide el enunciado, 100 mV de salida para un R_x conocido de 10 kΩ, la escala del óhmetro queda en 10 mV por cada 1000 Ω. Bajo esa escala un resistor de 1 MΩ tendría que dar 10 V a la salida.

El resistor bajo prueba se midió primero con el multímetro de laboratorio, que se toma como patrón, y dio $R_x = 0.9959\,\mathrm{M\Omega}$. Con ese mismo resistor conectado entre las terminales, la salida del circuito fue de $V_{out} = 14.63\,\mathrm{V}$, que llevada a resistencia con la escala calibrada equivale a una lectura de 1.463 MΩ.

$$\%\,\text{error} = \frac{1.463\,\mathrm{M\Omega} - 0.9959\,\mathrm{M\Omega}}{0.9959\,\mathrm{M\Omega}} \times 100$$
$$\%\,\text{error} = 46.9\,\%$$

El error es grande y además es un factor casi constante, la lectura sale 1.469 veces el valor patrón. El origen está en la calibración, ya que el montaje se hizo con un potenciómetro de 10 kΩ y el enunciado pide uno de 1.0 kΩ. Devolviendo la cuenta desde la salida medida se obtiene la tensión de entrada con la que realmente se trabajó.

$$V_{in} = \frac{V_{out}\,R_i}{R_x} = \frac{14.63\,\mathrm{V} \times 1\,\mathrm{k\Omega}}{995.9\,\mathrm{k\Omega}} = 14.69\,\mathrm{mV}$$

Son 14.69 mV en lugar de los 10 mV que corresponden a la escala de 10 mV/1000 Ω, y esa razón de 1.469 es justo el 46.9 % de error que dio la lectura. El potenciómetro más grande hace muy difícil acertarle al punto de ajuste. Con el de 1.0 kΩ la tensión de entrada recorre de 0 a 148 mV y los 10 mV de la calibración caen cerca del 7 % del recorrido del cursor, mientras que con el de 10 kΩ el rango sube hasta 1.36 V y esos mismos 10 mV quedan por debajo del 1 %, donde un movimiento mínimo corre toda la escala.

Vale agregar que a 14.63 V la salida ya queda cerca del límite de excursión del 741C, así que con esa calibración un resistor apenas mayor se habría recortado. Repitiendo la medición con el potenciómetro de 1.0 kΩ, la lectura para este R_x tendría que quedar alrededor de los 9.96 V esperados.